Zain Khan

software / mechatronics engineer

zainmk | zainmkhan | zainkhan.vercel

contact

zain.mk@hotmail.com
(403) 615 - 9897

education

Western University | 2016 - 2021

- Mechatronics Systems Engineering (BESc.)
- Computer Science (BSc.)

professional summary

A multidisciplinary software engineer capable of applying hardware/software solutions to automate or optimize data processes and flows, from it's collection to analysis.

I am currently seeking an opportunity that allows me to utilize my knowledge of automation & software development, alongside the next generation of AI tools.

work experience

September 2024 - Present
Calgary, AB

GCT Maintenance Solutions | *Software Engineer*

- Lead developer for a SaaS ERP platform built for the mining industry. Full stack development from the ReactJS frontend; to the RESTful NodeJS backend API server.
- Knex & SQL for schema design and database management, creating quick and efficient queries/mutations. Migration/Seed files for testing and database recovery plans.
- Establishing Azure Cloud infrastructure resources to facilitate application. Adhering to RBAC for authentication/authorization between services, users and developers. Terraform (IaC) to automate creating/managing infrastructure across multiple environments. Github Actions workflows to setup automating CI/CD flows across multiple environments.
- Frequent 'discovery' and PoC based projects to explore sustaining application development and iterating upon new emerging tools.
- Defining software design coding principles & practices. Leading a team of developers (interns) to effectively distribute, assign and complete tasks as per Agile principles.
- Developer, as contract, for *AgnicoEagle's JDEO ETL* rollout project; assisting with transitioning data from SAP ERP systems to JDE, for several Ontario-based mines.
- Planning, writing, documenting ETL flows for various resources with Pyspark (Databricks), SQL and authenticated Azure resources. Primarily responsible for transformation & filtering of data to meet business and schema requirements/specifications as well as loading data via orchestration APIs as defined per documentation.
- Auditing data points during transformation via assigned 'keys' to confirm ETL flows and documentation to explain ETL processes for final approval and handover to client.

August 2021 - June 2023
Toronto, ON

Heliolytics | *Frontend Developer*

- Lead developer for an internal analysis React based web tool, built to replace the legacy tooling used to analyze thermal images of solar panels and assess/assign deformities.
- Full-stack responsibilities include managing a GraphQL database, writing Python scripts, implementing/testing (PyTest) internal API endpoints, and utilizing ML and automation algorithms.
- Worked closely on developing a SaaS satellite imaging product, focused on providing visual assessments of solar array site management and developing.
- Responsible for defining processes between product, design and QA to ensure information is shared universally and consistently across teams.
- Documenting code design and infrastructure, to iterate effectively and communicate software designs and decisions into non technical vocabulary.

technical stack

- **FRONTEND |** ReactJS, TypeScript, HTML, CSS
- **BACKEND |** Python, Java, NodeJS
- **AI / ML |** TensorFlow, Keras, PyTorch, Numpy, Pandas, Pillow, Cursor IDE
- **EMBEDDED |** C, C++, MATLAB , Simulink
- **DATABASE |** SQL, MySQL, GraphQL, PostgreSQL, MongoDB, Knex, Pyspark
- **CLOUD |** Unix, Git, Kubernetes, Docker, Azure Cloud, Terraform, Github Actions, Databricks

certificates

MICROSOFT CERTIFIED: AZURE FUNDAMENTALS

February 20th, 2025

IBM AI ENGINEERING BY IBM ON COURSERA

August 8th, 2024

May 2020 - August 2020
Calgary, AB

University of Calgary | *Developer*

- Primary developer as part of a multidisciplinary team to assist in transitioning teaching resources online for the Chemical Engineering Department at the University of Calgary, during COVID-19.
- Utilized JavaScript, HTML and CSS to create interactive web pages consisting of educational content and calculators for students to use and a database to keep track of individual progress.

extracurricular & projects

ReactJS, Python

thumbs

- A web application allowing users to track 'stories' in the form of movies and games, while allowing ratings on a binary system. Utilizing the collected data to further predict and suggest new content.
- Based in ReactJS with Material UI components and a supporting FastAPI – Python backend to serve endpoints for managing computationally heavy and secure processes such as authorization and database security.

Java

LocalNeighborhoodCNN

- A locally implemented convolutional neural network, developed to understand the CNN architecture and how such a network operates to classify images efficiently. Trained and tested on the MNIST database.
- Programmed in Java from the ground up, including implementations for matrix operations, backpropagation calculus as well as the convolutional, max pool, and fully connected neural network layers.

Python

PyCOVID

- A script written to automatically collect COVID and environmental data to perform various ML analysis and predict trends of cases as well as possible influencing factors.
- Written in Python utilizing Selenium to automate web scraping publicly accessible data and sklearn to perform machine learning data analysis on the data collected.

Python, C++

WERC (Western Engineering Rocketry Club)

- Responsible for managing and implementing onboard control systems to collect telemetry information during the rocket's flight.
- Utilized a Raspberry Pi with C++ to collect data such as altitude and speed, while utilizing Python scripts to filter the data to a more usable form for analysis to improve on future flights

C++, MATLAB

MSE DESIGN PROJECT

- Mechatronics design project focused on creating a fully autonomous robot to perform a series of coordinated tasks incorporating aspects of electrical, mechanical and software design principles
- Utilized an onboard Arduino microcontroller programmed with C++. Troubleshooting the multidisciplinary aspects of the design, with the use of tools such as an oscilloscope, multimeter, and Matlab's Simulink to model transducers.